SHORT COMMUNICATION

Correlation of the patient's reported outcome Inflammatory-RODS with an objective metric in immune-mediated neuropathies

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Background and purpose: There is increasing interest in using patient-reported outcome measures (PROMs) in clinical studies to capture individual changes over time. However, PROMs have also been criticized because they are entirely subjective. Our objective was to examine the relationship between a subjective PROM and an objective outcome tool in patients with Guillain–Barré syndrome (GBS), chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) and gammopathy-related polyneuropathy (MGUSP).
Methods: The Inflammatory Rasch-built Overall Disability Scale (I-RODS©, a multi-item scale that examines functionality) was completed by 137 patients with newly diagnosed (or relapsing) GBS (55), CIDP (59) and MGUSP (23) who were serially examined (GBS/CIDP, T0/T1/T3/T6/T12 months; MGUSP, T0/T3/T12). Possible association between the I-RODS findings and the vigorimeter scores, an objective linear instrument to assess grip strength, was examined.
Results: A significant correlating trend was found between the I-RODS and grip strength scores for the overall group and in each illness, independently.
Conclusion: The objectivity of patients' subjective report on their functional state based on a strong correlation between the I-RODS and grip strength in patients with GBS, CIDP and MGUSP has been demonstrated. These findings provide further support to use the I-RODS and grip strength in future clinical studies in these conditions.

Introduction

Patient-reported outcome measures (PROMs) are used frequently as end-points in clinical trials and outcome studies [1,2]. PROMs are measurements of a patient's health condition obtained directly from the patient's own judgement, thus capturing their voice without input from a clinician. However, the use of these subjective outcome measures is controversial, and there are concerns about their validity, reliability and objectivity, as well as other factors [3–10]. One of the major criticisms for PROMs is their lack of objectivity, i.e. the ability to accurately reflect what they purport to measure and independently correlate with other unbiased outcome measures, and therefore many investigators have been reluctant to their use in clinical studies. Furthermore, there are virtually no studies that have systematically examined the objectiveness of a PROM in peripheral neuropathies.

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The Inflammatory Rasch-built Overall Disability Scale (I-RODS©) is a PROM that has been demonstrated to be the best surrogate outcome measure to assess activity limitations and participation restrictions in patients with Guillain–Barré syndrome (GBS), chronic inflammatory demyelinating polyneuropathy (CIDP) and immunoglobulin M monoclonal gammopathy of undetermined significance related polyneuropathy (MGUSP) [11–13]. The I-RODS represents the patient's selection of everyday activities reflecting general functionality. In contrast to the subjective I-RODS, the vigorimeter is an objective metric that assesses grip strength, and it has been demonstrated to be a precise linear measure that is highly sensitive in capturing changes over time in these patients, despite being criticized as not being reliable enough to assess responsiveness [14–17]. Our study was designed to examine the objectivity of patients' subjective reporting of their functional status in patients with inflammatory neuropathies [11,18]. It was hypothesized that the I-RODS would show a strong correlation with grip strength, thus demonstrating indirectly the objectivity of this PROM [15].

Methods

Patients, eligibility and ethical aspects

General characteristics, inclusion/exclusion criteria and ethical approval of the 137 patients enrolled have been published [13]. Briefly, 55 patients with GBS, 59 with CIDP and 23 with MGUSP with a newly diagnosed (or relapsing) neuropathy were examined prospectively over 12 months (Table 1). GBS and CIDP patients were examined at T0, T1, T3, T6 and T12 months, and patients with MGUSP at T0, T3 and T12 months, based on the indolent course of the latter illness. Patients were recruited and treated at the discretion of their physician as part of the Peripheral Neuropathy Outcome Measures Standardization (PeriNomS) study [19]. All patients were 18 years or older and met the international criteria for their illness [20–22]. Patients were excluded if they had any other potential cause for polyneuropathy, family history of neuropathy or exposure to neurotoxic medication or alcohol abuse [19].

Patients were recruited during hospital admission or at outpatient clinics at 26 international neuromuscular centers in the USA (Boston and Detroit), Canada (Ontario and Toronto), Brazil (Ribeirão Preto), UK (London), Italy (Milan, Rome and Venice), France (Paris and Marseille), Spain (Barcelona), Belgium (Brussels) and the Netherlands (Maastricht, Amsterdam, Utrecht and Rotterdam). The local medical ethics committee in each participating center approved the protocol. Written informed consent was obtained from all participants.

Table 1 Characteristics of the study population

	All patients, n = 137		
Illness	GBS (n = 55)	CIDP (n = 59)	MGUSP (n = 23)
Age			
Mean (SD)	53.4 (17.7)	52.7 (14.7)	62.8 (10.6)
Range	19–90	18–79	45–85
Gender, n (%)			
Female	18 (32.7)	17 (28.8)	8 (34.8)
Male	37 (67.3)	42 (71.2)	15 (65.2)
Years since diagnosis			
Mean (SD)	–	1.8 (3.1)	3.3 (5.3)
Range	–	0–16	0.2–25

GBS, Guillain–Barré syndrome; CIDP, chronic inflammatory demyelinating polyradiculoneuropathy; MGUSP, IgM monoclonal gammopathy of undetermined significance related polyneuropathy.

Study design and assessments

Patients' report on outcome

The I-RODS contains 24 activity and participation items graded from easy to difficult [able to read a newspaper/book (easiest item) up to being able to run (most difficult item)] (Table S1) [11]. Verbal and written instructions were provided to the researchers and patients on how to respond to the questions regarding the outcome measures. The I-RODS raw findings were subsequently transformed into Logits, the fixed unit used by Rasch analysis, before further data analyses took place [11].

Grip strength assessment

All researchers were trained on how to perform and record the findings of the vigori-dynamometer (0–160 kPa) in a standardized manner. Also, a picture-enriched training manual was provided that demonstrated the appropriate methods of data collection to each participating institute as part of the PeriNomS study [19]. Grip strength was assessed at each visit three times for both hands in an alternating order with 30 s rest in between, and the mean of the assessments for each hand was calculated.

Statistics

The correlation between the subjective I-RODS and the objective grip strength was examined using random effects linear regression analyses, taking into account the association of the data caused by the longitudinal structure. The latter was achieved using the program xtreg in Stata 13.0 for Windows XP (Breda,

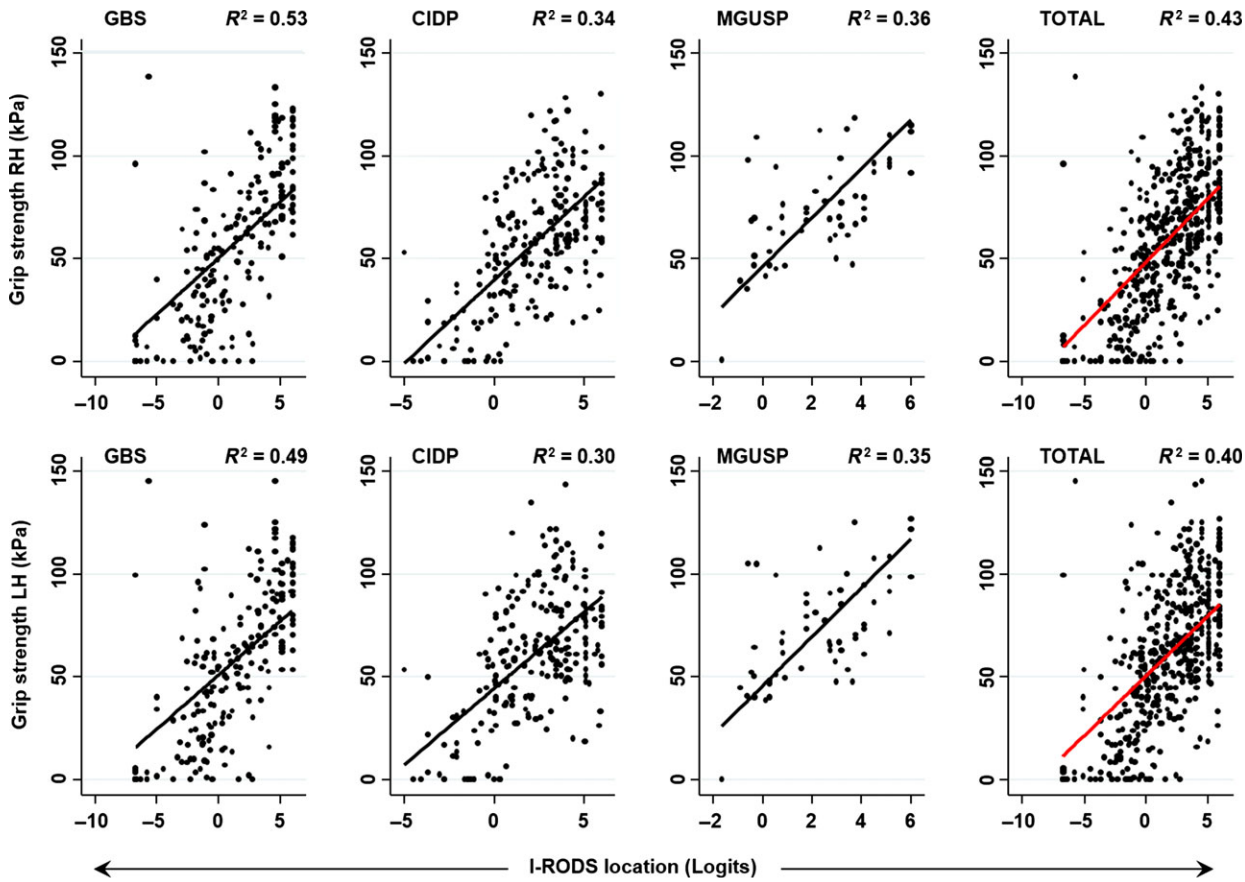

Figure 1 I-RODS© in correlation with grip strength. Top panels right hand (RH), bottom panels left hand (LH). I-RODS©, Inflammatory Rasch-built Overall Disability Scale; GBS, Guillain–Barré syndrome (n = 55 patients); CIDP, chronic inflammatory demyelinating polyradiculoneuropathy (n = 59 patients); MGUSP, IgM monoclonal gammopathy of undetermined significance related polyneuropathy (n = 23 patients). All patients were examined during 12 months; the data obtained for the assessments were pooled and used for correlation purposes. The association between the two variables was determined using regression studies, taking into account the clustering of data at the individual patient's level in time [23], and was expressed as R^2: the fraction of variance explained by the independent variable (changes in I-RODS) on the dependent variable (grip strength) + time factor from the regression model; for all associations, $P \leq 0.001$.

Netherlands), which is based upon a time-series regression model as described by Dwyer *et al.* [23]. The strength of association between the variables was expressed as R^2: the fraction of variance explained by the independent variable (changes in I-RODS) on the dependent variable (grip strength) + time factor from the regression model. All analyses were undertaken using Stata 13.0 for Windows.

Results

General characteristics

Patients' characteristics are presented in Table 1. Most patients received treatment (93% in GBS, 71% in CIDP and 50% in MGUSP). Intravenous immunoglobulin (IVIg) was prescribed most commonly (82% in GBS, 53% in CIDP and 35% in MGUSP); corticosteroids were used either in combination with IVIg or alone (15% in GBS and 15% in CIDP). Plasma exchange was performed in nine GBS cases (16%), three CIDP (5%) and four MGUSP cases (17%). A small number were treated with an immunosuppressant agent (two cases with MGUSP received cyclophosphamide; one patient with MGUSP was treated with azathioprine). A total of 600 patient visits were completed over 12 months follow-up. The data obtained at these visits were pooled to determine the correlation findings between the grip strength and I-RODS.

Grip strength in relation to I-RODS

Both assessments had acceptable normal distribution pattern. There was a significant correlation between grip strength values and the functional categories of the I-RODS (Fig. 1). The correlation was most clear for

GBS followed by CIDP and MGUSP patients. Quite similar patterns were seen between right and left hand grip strength scores versus I-RODS categories (Fig. 1).

Discussion

In this study it has been demonstrated that a patients' own (subjective) personal report on executing daily/social activities (as captured with the I-RODS) showed a significant correlation with grip strength as a linear objective measure in patients with GBS, CIDP and MGUSP. No difference was seen between the right and left hand grip strength values related to the subjective measures completed by these patients. The use of grip strength in CIDP (and comparable conditions) has been questioned as an appropriate tool because it does not address lower limb function and therefore may be unreliable as a measure of treatment response [17]. Grip strength was therefore addressed as being 'imperfect' in daily practice in these conditions [17]. Our findings show a significant correlation between grip strength and I-RODS, which also contains daily and social activities that are mobility driven, such as walking one flight of stairs, travelling by public transportation, dancing, standing for hours, and ability to run. The correlation found, although not surprising to physicians treating patients with GBS, CIDP and MGUSP, also indicates that the functional limitations have a more general impact, extending to the upper limbs as well.

Grip strength has been shown to be a sensitive tool to detect clinically relevant improvement or deterioration, sometimes with greater sensitivity than other commonly used generic disability measures [14]. Its scientific requirements in inflammatory neuropathies have been thoroughly demonstrated [14–16,24]. In other conditions, grip strength is considered a good prognostic indicator of functional recovery [25,26]. This easily applicable tool provides much greater objective information regarding a patient's general functioning and well-being, despite only measuring distal upper limb strength, and explains in CIDP and MGUSP up to a third of functional deficit in these conditions, and half of the daily/social activities encountered by patients with GBS are explained by grip strength values (Fig. 1). In addition, the I-RODS is the only interval measure to assess general functionality in patients with GBS, CIDP and MGUSP and has fulfilled all modern Rasch-built scientific requirements [11,13,27]. Its superiority over generally applied often ordinal-based impairment and disability measures has recently been demonstrated, as well as showing some cross-cultural validity [13,28]. Through a consensus meeting with international experts in the field of inflammatory neuropathies and patients' representatives, it was concluded that the I-RODS and the vigorimeter are the most sensitive tools in these conditions [12].

In future perspectives for daily clinical practice, the I-RODS and grip strength could be used to help establish patients' 'baseline' performance and to evaluate the effects of treatments such as immunomodulatory therapy. Currently efforts are being undertaken to make the I-RODS more easily and widely accessible through a website and App construction, as well as embedding it into registries in inflammatory neuropathies [29]. Our findings support, in addition to the literature findings, the routine use of the I-RODS and grip strength in clinical trials and follow-up studies in patients with inflammatory neuropathies.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. List of activities as part of the I-RODS©

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Appendix

The PeriNomS Study Group

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